Static and kinetic friction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish adjustable static friction from sliding friction.

SUCCESS CRITERIA - I can...

1. Compare a push with the static-friction limit.
2. Find actual friction for rest or sliding.
3. Explain friction direction using relative slipping.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A runner pushes the ground backward with 50 N. Name the partner force, its size and the object it acts on. Explain its role in the runner's motion.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $0 \le f_s \le \mu_s N$; $f_k = \mu_k N$.
s = static; k = kinetic; N = normal force. Static friction adjusts up to its maximum. Kinetic friction applies during sliding.

Worked reasoning: A box remains still while pushed with 3 N. Static friction is 3 N opposite the push, provided its maximum is at least 3 N. Static friction need not equal its maximum.

Your turn: Draw the horizontal forces. Then redraw for a 5 N push that is still below the maximum.

Marzano check - circle: ① ② ③

03 Below the limit

The box stays at rest with an 8 N push right. Maximum static friction is 15 N. Find actual friction and net force.



The dot represents the object. At rest; static limit 15 N

Marzano check - circle: ① ② ③

04 Crossing the limit

For the same box, the push increases to 18 N. Sliding friction is 12 N. Once it slides right, find net force.

Marzano check - circle: ① ② ③

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05 Can friction help?

Your foot pushes backward on the ground without slipping. Which way does ground friction act on your foot? Explain why“friction always points backward”is wrong.

Marzano check - circle: 1 2 3

06 Choose a method

A cart starts at 4 m and moves right at 6 m/s for 3 s. Predict final position. Allowed endpoints are 20 to 24 m; state this interval and one assumption.

Marzano check - circle: 1 2 3

07 Exit check

Maximum static friction is 20 N.A 12 N push does not move the box. Find actual friction. A 25 N push starts sliding with 16 N kinetic friction: find net force.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A box has maximum static friction 8 N and kinetic friction 6 N. Predict the behavior for pushes of 4, 8 and 10 N. Draw the sliding case.

Marzano check - circle: 1 2 3